Metric

DOD-W-70666 (AR)
15 December 1986

MILITARY SPECIFICATION

WARHEAD, GUIDED MISSILE, HE, M207E6 TRUMPET LINER FOR

This specification is approved for use within the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the quality assurance provisions and special requirements not covered by the drawings for the fabrication and packaging of the liner for M207E6 Warhead.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition Standard Quality Assurance Provisions, General Specification for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament, Munitions and Chemical Command, Attn: AMSMC-QA, Dover, New Jersey 07801-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1336

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components

2.1.1 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

Drawings

U.S. ARMY ARMAMENT, RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

12589943 LINER, Trumpet Precision TOW

(Copies of specifications, standards, handbooks, drawings publications and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adapted shall be those listed in the issues of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue in the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM-B-133 / ASTM-B-133M Specification for Copper Rod, Bar, and Shapes

ASTM-B-187 Specification for Copper Bus Rod, Bar, and Shapes

ASTM-B-249 / ASTM-B-249M Specification for General Requirements for Wrought Copper and Copper-Alloy Rod, Bar, and Shapes

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ASTM-E-18	Rockwell Hardness and Rockwell Superficial Hardness of Maetallic Materials, Test for
ASTM-E-92	Vickers Hardness of Metallic Materials
ASTM-E-112	Standard Methods for Determining the Average Grain Size
ASTM-F-68	Oxygen Free Copper in Wrought Forms for Electron Devices

(ASTM Standards for test methods are generally available in reference libraries. They are also distributed among technical groups and certain federal agencies. Application for copies should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadelphia, Pennsylvania 19102)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS.

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Parts. The parts shall comply with all requirements specified on drawing 12589943, all associated drawings and with the requirements of applicable standards and specifications.

3.3 Starting stock. Raw material to be introduced into the cold forged process shall be analyzed for mechanical, electrical, physical, and chemical properties as shown below.

3.3.1 Physical properties of starting stock. The sample of starting stock shall comply with the mechanical and electrical requirements of Copper number 101 (C10100), Temper 060 as specified by Table 2 of ASTM-B-133 (ASTM-B-133M) or Table I of ASTM-B-187, with class 1 or 2 material acceptable.

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3.3.2 Chemical composition of starting stock. The sample of starting stock shall comply with the chemical requirements of Copper number 101 (C10100) Temper 060 as specified by ASTM-B-133 (ASTM-B-133M) or ASTM-B-187, supplemented by Table I of ASTM-F-68, with class 1 or 2 material acceptable.

3.4 Finished liner. The finished liner shall meet the grain size and hardness requirements as specified on the drawing.

3.5 Profile. The finished liner shall meet the profile requirement as specified on the drawing.

3.6 Data requirements. The contractor shall generate data in accordance with the requirements of the data item description cited in 6.4.

3.7 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.8 Process control

3.8.1 Process documentation. Concurrent with submission of the first article sample (see 4.3), the contractor shall submit a written description of the process used to produce the samples together with the means by which the process can be verified in production. The method of verification shall be reviewed by and must be acceptable to the government. See Data Item Description DI-P-1604 (Tailored). Proposed changes to the process which produced the accepted first article sample shall be submitted to the government prior to actual implementation of the change. A new first article sample will be required for each process change if determined necessary by the technical agency (see 6.5, if non U.S. manufacturer).

3.8.2 Liner traceability. The contractor shall also submit a method to establish liner traceability with specified lots. This information shall be furnished as a part of DI-P-1604 (Tailored) above and shall be reviewed by and must be acceptable to the government.

3.9 Workmanship. All liners shall be fabricated in a thorough, workmanlike manner. The parts shall be free of burrs, chips, sharp edges, cracks, unblended radii, dents, tool marks, chipped edges, warpage, gouges, surface defects, dirt, grease, corrosion products, burn marks, and other foreign matter. The cleaning method used shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspections. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made part of this specification.

4.1.1 Responsibility for compliance. All items must meet all the requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Certification provisions. Certification shall include test data and results for each characteristic specified below. Certification shall be required for all material used in manufacturing the liners and prior to performing final acceptance inspection. Certification of Conformance shall include name of raw material producer, date of purchase, purchase order number, quantity, and all chemical, mechanical, and physical analyses performed on the material. The following statement shall be included:

"The undersigned, individually, and as the authorized representative of the contractor, warrants and represents that: All the information supplied above is true and accurate; the material covered by this certificate conforms to all contract requirements (including but not limited to the drawings and specifications); the inspection and test results, and the analyses appearing herein are true and accurate; and this certificate is made for the purpose of inducing payment and with knowledge that the information and certification may be used as a basis for payment."

CHARACTERISTIC	TEST DATA TO COMPLY WITH
Raw material	3.3. 3.3.1 and 3.3.2

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4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First article inspection (see 4.3)

b. Quality conformance inspection (see 4.4)

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the contracting officer, for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following item, in the sample quantity indicated.

PART DESCRIPTION	DRAWING	QUANTITY
LINER, Trumpet Precision TOW	12589943	20

4.3.2 Inspections to be performed. Examine and test in the order listed on Table I and Figure 1. Sample material of the starting stock shall be selected prior to fabrication and used for hardness and chemical composition tests. Examine all twenty (20) liners for major characteristics 103 - 112 and all minor characters of 4.4.2.1. Inspections for characteristics 113 and 114 of 4.4.2.1, shall be performed on a sample of ten (10) liners, randomly selected from the original sample of twenty (20) liners. Inspections for characteristics 115 and 116 of 4.4.2.1, shall be performed on a sample of ten (10) liners; five (5) of which shall be selected from the sample designated for profile tests and five (5) from the remaining ten (10) first article samples (see Figure 1). A certificate of conformance for raw material, is also required.

4.3.3 Rejection. See MIL-A-48078.

4.3.4 Completion. Upon completion of first article testing all piece parts shall be properly labeled/identified and sent to: Commander, U.S. Army Armament, Research, Development, and Engineering Center, Attn: SMCAR-CCH-A, Dover, N.J. 07801-5001.

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Table I - First article inspection

CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
	Liner, Trumpet Precision TOW				12589943
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Physical properties of starting stock	Material Sample		3.3.1	4.5.2.1.1
	Chemical composition of starting stock	Material Sample		3.3.2	4.5.2.1.2
	Examination of Defects (see note 1)	20		3.2	4.4.2.1
	Profile of inner surface (see note 2)	10		3.5	4.6
	Profile of outer surface (see note 2)	10		3.5	4.6
	Hardness of finished liner (see note 3)	10		3.4	4.5.2.2.1
	Grain size of finished liner (see note 3)	10		3.4	4.5.2.2.2
	See Figure 1				
	*Examine and test in the order listed.				

NOTES:

Note 1: Major characteristics 103 through 114 and all minor characteristics.

Note 2: Same 10 liners shall be used for inner and outer surface profile tests.

Note 3: Same ten (10) liners shall be used for hardness and grain size tests. Five (5) trumpet liners samples will be taken from those utilized for profile testing and five (5) from the remaining ten (10) liners.

AMSMC Form 1570a, 1 Apr 85 Replaces DRDAR-QA Form 160a, 1 Jun 83, which may not be used.

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First Article Test Procedure for TOW Trumpet Liner

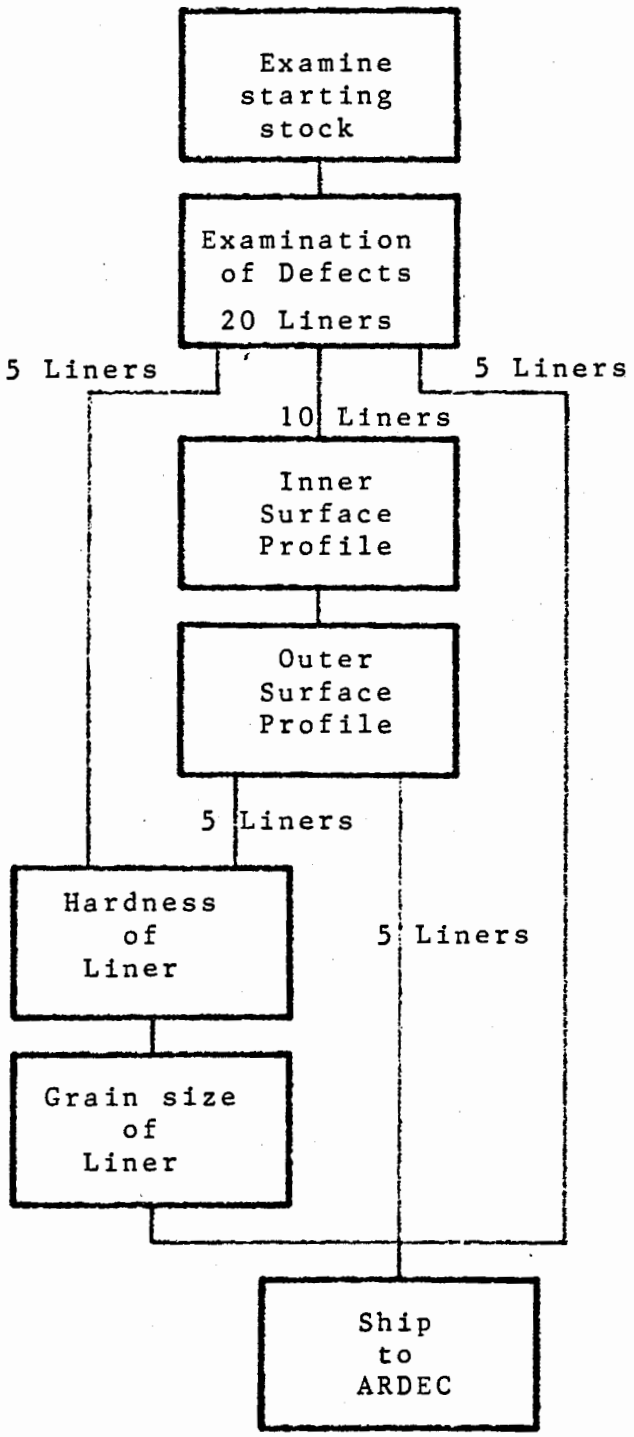


Figure 1

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall be defined as liners made from the same heat of copper, subjected to the same manufacturing process using the same forging dies, during one (1) month's production. Lot formation shall comply with applicable provisions of MIL-A-48078.

4.4.2 Examinations. See MIL-A-48078. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Liner, Trumpet Precision TOW	1 OF 2	12589943
			NEXT HIGHER ASSEMBLY 12589944

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Physical properties of starting stock	Material sample		3.3.1	4.5.2.1.1
102	Chemical composition of starting stock	Material sample		3.3.2	4.5.2.1.2
103	Wall thickness at datum A/C (see note 5 on dwg.)		0.40%	3.2	Gage
104	Wall thickness (see note 5 on dwg.)		0.40%	3.2	Gage
105	Wall thickness at datum B/D (see note 5 on dwg.)		0.40%	3.2	Gage
106	Length from base to datum A		0.40%	3.2	Gage
107	Length from base to datum C		0.40%	3.2	Gage
108	Length from base to inner apex		0.40%	3.2	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

DOD-W-70666 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Liner, Trumpet Prevision TOW	2 OF 2	12589943
			NEXT HIGHER ASSEMBLY 12589944

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
MAJOR					
109	Overall length		0.40%	3.2	Gage
110	Outside diameter		0.40%	3.2	Gage
111	Runout of base		0.40%	3.2	Gage
112	Length of liner step		0.40%	3.2	Gage
113	Profile of inner surface (see note 1)	7		3.5	4.6
114	Profile of outer surface (see note 1)	7		3.5	4.6
115	Hardness of finished liner (see note 1) (see note 6 on dwg.)	7		3.4	4.4.3.1/4.5.2.2.1
116	Grain size of finished liner(see note 1) (see note 6 on dwg.)	7		3.4	4.4.3.2/4.5.2.2.2
MINOR					
201	Presence of four (4) notches on liner step		0.65%	3.2	Visual
202	Surface finish (see note 4 on dwg.)		0.65%	3.2	Gage
203	No discontinuity of machined or forged surfaces (see note 12 on dwg.)		100%	3.2	Visual
204	Workmanship		100%	3.9	Visual

NOTES:

Note 1: Fourteen (14) liners shall be used for lot acceptance testing. Seven (7) shall be used for profile inspections and the other seven (7) for hardness and grain size test.

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4.4.3 Lot acceptance testing. Testing is described in Quality Conformance Inspection tables. A sample of 14 trumpet liners shall be utilized for lot acceptance tests.

4.4.3.1 Profile sampling. Perform and record inner surface and outer surface profile measurements on seven (7) liners to be designated for penetration tests.

4.4.3.2 Hardness and grain size sampling. Randomly select two (2) liners from the first 25 liners manufactured, two (2) liners from the last 25 liners manufactured, and three (3) liners from the remainder of the lot for a total of seven (7) liners and perform the destructive tests in accordance with 4.5.2.2.1 and 4.5.2.2.2.

4.4.4 Inspection equipment. The inspection equipment required to perform the examination and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Methods of inspection

4.5.1 Materials, components and processes. Compliance with all requirements of Section 3 of this specification shall be ascertained by current and continuing examination of inspection and test data to determine that all components (parts and materials) have been inspected and tested and found to comply with their respective drawing and specification requirements, and that all specified manufacturing processes have been followed.

4.5.1.1 Visual and mechanical inspection. The visual, manual and mechanical inspections and tests shall verify compliance with the requirements of Section 3 and 5 of this specification in accordance with Table I and 4.4.2 herein.

4.5.2 Metallurgical and chemical tests. The following tests shall be performed using equipment and procedures prescribed by ASTM-B-249 (ASTM-B-249M).

4.5.2.1 Starting stock

4.5.2.1.1 Physical properties of starting stock. Hardness tests shall be per ASTM-E-18. Failure of the sample to comply with 3.3.2 shall cause rejection of that quantity of starting stock represented by the sample. These test data shall be independent from the supplier data. Results of test data shall be submitted in accordance with DI-T-5247 (tailored) (see 6.4).

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4.5.2.1.2 Chemical composition of starting stock. Chemical composition tests shall utilize methods prescribed by ASTM-F-68. Failure of the sample to comply with the requirements of ASTM-B-187, supplemented by ASTM-F-68, shall cause rejection of that quantity of starting stock represented by the sample. These test data shall be independent from supplier data. Results of test data shall be submitted in accordance with DI-T-5247 (tailored) (see 6.4).

4.5.2.2 Finished liner.

4.5.2.2.1 Hardness of finished liner. Hardness tests of finished liners shall be per ASTM-E-92. The sample liners shall be sectioned longitudinally into two (2) equal parts and marked for exact location relative to the finished liner. The Vickers hardness readings shall be taken at the cross-section and inner surface on both sections of the liner (see Figure 2). A total of 22 readings shall be identified as to their location on the liner. The recorded readings shall be analyzed for variations in hardness in the longitudinal and circumferential directions. If any of the sample liners do not meet the hardness requirement specified on the drawing, the lot represented by the samples shall be rejected. The hardness test data shall be submitted in accordance with DI-T-5247 (tailored) (see 6.4). This is a destructive test.

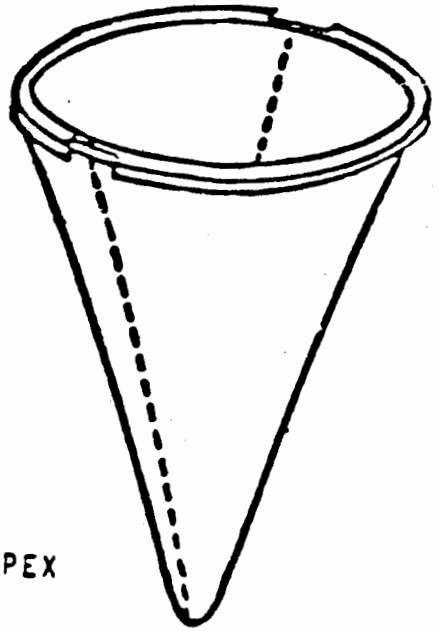
a. The cross-sectioned surface hardness readings shall be taken one and a half (1 1/2) inches (3.81 cm) from the base and every two (2) inches (5.08 cm) thereafter in a longitudinal direction, from base to apex. Only one section shall be inspected but at each cross-section end for a total of eight (8) readings.

b. The inner surface hardness reading shall be taken one (1) inch (2.54 cm) from the base and every one (1) inch (2.54 cm) thereafter in a longitudinal direction, from base to apex, on the longitudinal centerline of each section for a total of 14 readings.

4.5.2.2.2 Grain size of finished liner. Grain size measurements of finished liners shall be per ASTM-E-112. One (1) test section from 4.5.2.2.1 shall be utilized for taking grain size. Grain size shall be taken at one and a half (1 1/2) inches (3.81 cm) from the base and every two (2) inches (5.08 cm) thereafter in a longitudinal direction, from base to apex, for a total of four (4) readings per sample liner (see Figure 2). If any of the sample liners do not meet the grain size requirements as specified on the drawing, the liner heat treat lot represented by the sample shall be rejected. The grain size test data shall be submitted in accordance with DI-T-5247 (tailored) (see 6.4).

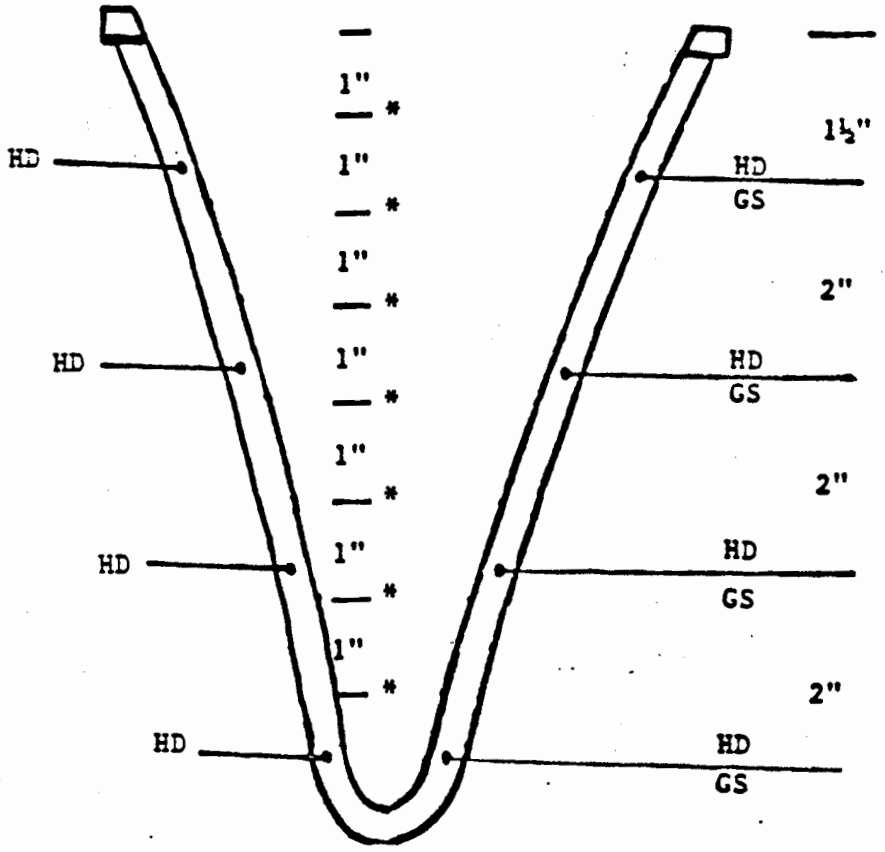
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SCHEMATIC OF SECTIONED TRUMPET LINER



Section liner into
two (2) equal sections

Location of Grain size (GS)
and Hardness (HD) tests



Center line hardness test (*) to be performed on both sections

FIGURE 2

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4.6 Profile. The sample liners shall have their profiles inspected using approved measuring and test equipment. If a sample liner does not meet a profile requirement, the lot shall be rejected. If all seven (7) lines are acceptable then identify them for penetration testing. (see 3.5)

5. PACKAGING

5.1 Packaging and packing. The parts shall be packaged and packed in accordance with the requirements of MIL-STD-1169.

6. NOTES

6.1 Intended use. The liner covered by this specification is a component of the TOW Warhead Section, Guided Missile, HE, M207E6.

6.2 Ordering data. See MIL-A-48078.

6.3 Submission of inspection equipment for design approvals. See MIL-A-48078. Submit designs as required to Commander, US Army Armament Research Development and Engineering Center, (ARDEC) ATTN: AMSMC-QAT-I, Dover, NEw Jersey 07801-5001.

6.4 Data requirements - Deliverable data required by the specification is cited in the following paragraphs:

Paragraph	Data Requirement	Applicable DID
3.5	Ammunition Data	DI-MISC-80043
3.8, 3.8.1, & 3.8.2	Manufacturing Methods Report	DI-P-1604 (Tailored)
4.5.2.1.1, 4.5.2.1.2, 4.5.2.2.1, & 4.5.2.2.2	Test Report	DI-T-5247 (Tailored)

The above data items shall be delivered to the following to Commander, U.S. Army ARDEC, Attn: AMSMC-QAT-B (D), Dover, N.J. 07801-5001.

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6.5 Non U.S. Manufacturer. If an OCONUS (non U.S. manufacturer) subvendor/supplier controls the process and considers it proprietary, then the following procedure may be used in place of 3.8.1. Contractor is authorized to retain process description, verification procedures, etc. at his facility in lieu of forwarding to this command. Personnel from this command consisting of the Product Quality Manager, a Quality Engineer, a Production Engineer, and a U.S. representative and their supervisor will be the only personnel authorized to view the Process Control Document etc., at the contractor's facility (names to be supplied upon request). The Government personnel do not have the authority to reproduce the document in whole or any part thereof (Note: they are also bound under U.S. Army Laws and Regulations not to reveal said document or any portion thereof, to any other person, either within or outside our Government agencies). The approval/disapproval (note: disapproval will only be for lack of information not method, technical procedures, etc.) of the document will be accomplished on site. Also, at all times said document will be under the cognizance of the contractor (i.e. the subvendor/supplier). It is noted that the prime contractor may arrange whatever he considers necessary in regard to his viewing of the Process Control Document. However, it is incumbent upon him to have in his contract with his subvendor/supplier's permission for Government viewing and verification of the process. The above permission must apply even if the prime contractor is denied viewing rights, etc. Note: this exception to 3.8.1 applies only to OCONUS subvendor/supplies. It does not apply to OCONUS prime or CONUS (U.S.) contractors whether prime or subvendor/supplier.

6.6 Subject term (Key Word) listing.

Liner, Trumpet Precision TOW
American Society for Testing and Material (ASTM)
Data Item Description (DID)
Starting stock
Grain size
Hardness
Process control
Traceability
Batch

6.5 Contractor. Unless otherwise specified, the word "Contractor," as used throughout this specification, shall mean any supplier or producer of items or material to the Government. This shall include commercial contractors, sub-contractors, Government-Owned, Contractor-Operated (GOCO) and Government-Owned, Government-Operated (GOGO) plants.

Custodian
Army-AR

Preparing activity
Army-AR
(Project 1336-A538)

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